

Students were asked to bring items from home to serve as examples for different types of reactions. The instructor provided additional reactants, solvents, heat, or a catalyst to demonstrate reactions with the household items.

The students brought aluminum foil, table salt, sugar, baking soda, vinegar, toilet bowl cleaner, hydrogen peroxide, and antacid tablets.

Experiment 1

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Table 1 Reaction Conditions and Visual Observations

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Table salt: NaCl	Dissolved in water and then added to a solution of 0.05 M AgNO_3	White solid precipitate formed
Vinegar: $\text{HC}_2\text{H}_3\text{O}_2(\text{aq})$	Added baking soda (NaHCO_3) to 2 mL vinegar in a test tube	Fizzed and bubbled; produced a gas
Toilet bowl cleaner: HCl(aq)	Added a small piece of Al foil into a test tube containing 2 mL of cleaner	After several minutes, bubbles appeared; bubbles were produced until the Al foil disappeared
Hydrogen peroxide: $\text{H}_2\text{O}_2(\text{aq})$	Some MnO_2 catalyst was added to H_2O_2 in a tall flask	Bubbles formed on the surface of the liquid

Experiment 2

The reactions that were observed to produce gas bubbles in Experiment 1 were repeated, and samples of the gases produced were collected. Each gas sample was tested by holding a burning (glowing) wooden splint within the reaction vessel to determine if the reaction produced O_2 , H_2 , or CO_2 . A glowing splint placed in O_2 will burn more brightly, one placed in H_2 will ignite the gas and cause a loud noise, and one placed in CO_2 will be extinguished. The results of the gas tests are recorded in Table 2.

Table 2 Glowing Splint Gas Test Observations

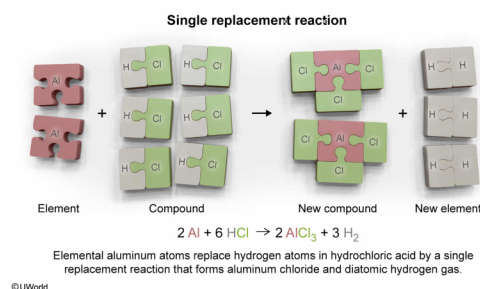
Household item	Reaction conditions	Glowing splint observations
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Hydrogen peroxide: $\text{H}_2\text{O}_2(\text{aq})$	MnO_2 catalyst and H_2O_2 were combined in a tall flask	Burned more brightly when placed in the gas

The reaction between the toilet bowl cleaner and the aluminum foil forms aqueous AlCl_3 in addition to the gas that is detected. Which of the following best explains this outcome of the reaction?

- ☐ A. Aluminum undergoes a double replacement reaction with the aqueous HCl in the cleaner. [3%]
- ✓ ☒ B. Aluminum participates in a single replacement reaction with the aqueous HCl in the cleaner. [83%]
- ☐ C. The AlCl_3 forms by a combination reaction between Al and the aqueous HCl in the cleaner. [8%]
- ☐ D. The AlCl_3 forms by a decomposition reaction between Al and the aqueous HCl in the cleaner. [4%]

Correct

83% answered correctly

Explanation:

Chemical reactions are classified into different types based on certain patterns that they display, which helps in predicting the products of reactions. The five common **types** of chemical reactions are single replacement, double replacement, combination, decomposition, and combustion.

A **single replacement** reaction involves the **replacement** of **one type of atom** with another in a reaction between a **compound** and a **neutral element** in its stable form. A neutral element that is sufficiently reactive loses or gains electrons and takes the place of one of the atoms in the compound. The atom that it replaces then loses or gains electrons to become a neutral element.

In the reaction discussed in this question, elemental aluminum $\text{Al}(\text{s})$ replaces hydrogen in HCl to form AlCl_3 . In turn, the hydrogen atoms combine to form elemental hydrogen gas (as indicated in Table 2), which exists as H_2 because elemental hydrogen is most stable as a neutral diatomic molecule.

Therefore, the reaction between $\text{Al}(\text{s})$ and HCl is a single replacement reaction. Note that although aluminum foil is protected by an oxide surface layer (Al_2O_3), HCl reacts with Al_2O_3 until it reaches the unprotected aluminum layer. At that point, the described reaction between $\text{Al}(\text{s})$ and HCl occurs.

(Choice A) The reaction is not a **double replacement** reaction because two ions do not exchange their respective counter-ions.

(Choice C) The reaction is not a **combination reaction** because two elements do not combine to form one new stable compound.

(Choice D) The reaction is not a **decomposition reaction** because a compound does not break down into two or more compounds or elements.

Educational objective:

Single replacement reactions involve the replacement of one type of atom with another in a reaction

between a neutral element and a compound such that a new compound and a new neutral element are produced.

Time spent: 0 sec

Subject:
General Chemistry

QID: 402067

System:
Interactions of Chemical Substances

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Vinegar: $\text{HC}_2\text{H}_3\text{O}_2(aq)$	Added baking soda (NaHCO_3) to 2 mL vinegar in a test tube	Fizzed and bubbled; produced a gas
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Hydrogen peroxide: $\text{H}_2\text{O}_2(aq)$	Some MnO_2 catalyst was added to H_2O_2 in a tall flask	Bubbles formed on the surface of the liquid

Experiment 2

The reactions that were observed to produce gas bubbles in Experiment 1 were repeated, and samples of the gases produced were collected. Each gas sample was tested by holding a burning (glowing) wooden splint within the reaction vessel to determine if the reaction produced O_2 , H_2 , or CO_2 . A glowing splint placed in O_2 will burn more brightly, one placed in H_2 will ignite the gas and cause a loud noise, and one placed in CO_2 will be extinguished. The results of the gas tests are recorded in Table 2.

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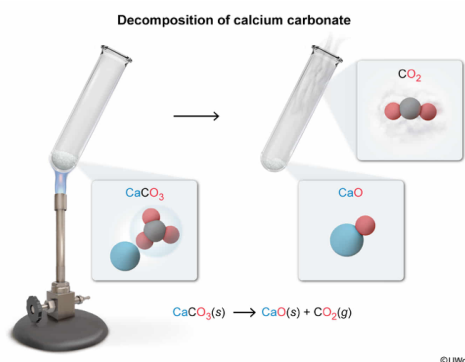
Which of the following is the reaction for the decomposition of calcium carbonate based on the information in Tables 1 and 2?

- ☐ A. $\text{CaCO}_3(s) \rightarrow \text{Ca}(s) + \text{CO}(g) + \text{O}_2(g)$ [7%]
- ☐ B. $\text{CaCO}_3(s) + 2 \text{HCl}(aq) \rightarrow \text{CaCl}_2(aq) + \text{H}_2\text{O}(l) + \text{CO}_2(g)$ [3%]
- ✓ ☒ C. $\text{CaCO}_3(s) \rightarrow \text{CaO}(s) + \text{CO}_2(g)$ [88%]
- ☐ D. $\text{CaCl}_2(aq) + \text{Na}_2\text{CO}_3(aq) \rightarrow \text{CaCO}_3(s) + 2 \text{NaCl}(aq)$

Correct

87% answered correctly

Explanation:



An important reaction type is the **decomposition reaction**, which involves the **breaking down** of a compound into **simpler, stable compounds** or into its **elemental constituents**. This type of reaction usually requires the addition of **energy** (such as heat) to break the bonds of the original compound.

In this question, calcium carbonate is isolated from the inactive ingredients in an antacid tablet and then placed over a flame. The gas produced is CO_2 , based on the results from the glowing splint test. This means that the white solid that remains is a new compound that does not contain carbon. Calcium preferentially forms a metal oxide in the presence of oxygen, so the other product must be CaO .

(Choice A) Although this is a decomposition reaction (one reactant), $\text{O}_2(\text{g})$ cannot be a product because the students observed that the glowing splint was *extinguished* when placed in the presence of the gas. This indicates that $\text{CO}_2(\text{g})$ must be a product.

(Choice B) According to Table 1, only heat was added to CaCO_3 , not HCl . Furthermore, the reaction between CaCO_3 and HCl is not a decomposition reaction because there are two reactants. Reactions with acids or bases are often **double replacement** reactions in which water is produced.

(Choice D) The question asks for the decomposition of calcium carbonate, which means CaCO_3 must be the only reactant (ie, on the left side) of the chemical equation.

Educational objective:

A decomposition reaction is a type of reaction in which energy is added to break down a compound into simpler, stable compounds or into its elemental constituents.

Time spent: 0 sec

Subject:
General Chemistry

QID: 402068

System:
Interactions of Chemical Substances

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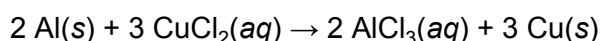
Experiment 2

The reactions that were observed to produce gas bubbles in Experiment 1 were repeated, and samples of the gases produced were collected. Each gas sample was tested by holding a burning (glowing) wooden splint within the reaction vessel to determine if the reaction produced O_2 , H_2 , or CO_2 . A glowing splint placed in O_2 will burn more brightly, one placed in H_2 will ignite the gas and cause a loud noise, and one placed in CO_2 will be extinguished. The results of the gas tests are recorded in Table 2.

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When 3.0 g of aluminum foil is added to 500. mL of the CuCl_2 solution indicated in Table 1, the following reaction takes place:



The limiting reactant is:

- ✓ ☒ A. aluminum. [75%]
☐ B. copper. [2%]
☐ C. copper(II) chloride. [21%]
☐ D. aluminum chloride.

Correct

75% answered correctly

Explanation:

Determining the limiting reactant

$2 \text{Al(s)} + 3 \text{CuCl}_2(\text{aq}) \longrightarrow 2 \text{AlCl}_3(\text{aq}) + 3 \text{Cu(s)}$

Mass Al	Molar mass	Mole ratio from reaction	Moles AlCl ₃ produced
3.0 g Al(s)	$\times \frac{1 \text{ mol Al}}{27.0 \text{ g Al}}$	$\times \frac{2 \text{ mol AlCl}_3}{2 \text{ mol Al}}$	= 0.11 mol
Volume CuCl ₂ (aq)	CuCl ₂ solution molarity	Mole ratio from reaction	Moles AlCl ₃ produced
0.500 L CuCl ₂ (aq)	$\times \frac{1.0 \text{ mol CuCl}_2}{1 \text{ L solution}}$	$\times \frac{2 \text{ mol AlCl}_3}{3 \text{ mol CuCl}_2}$	= 0.33 mol

0.11 mol < 0.33 mol

- Al produces the smallest theoretical amount of product
- Al is the limiting reagent

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During a chemical reaction, the reaction will continue until one of the reactants is fully consumed. The **limiting reactant** is the reactant that is completely consumed first, thus stopping the reaction and limiting the **amount of product** that can be produced.

The limiting reactant can be found by determining the number of moles of product that the stated amounts of each reactant could theoretically form. This is based on the number of moles of each reactant and the **ratios** of the **stoichiometric coefficients** in the **balanced** reaction equation. The limiting reactant is the one that yields the smallest amount of product.

Based on the **molar mass** of aluminum, a 3.0 g sample of aluminum foil is equivalent to 0.11 moles of aluminum.

$$3.0 \text{ g Al} \times \frac{1 \text{ mol Al}}{27 \text{ g Al}} = 0.11 \text{ mol Al}$$

According to the balanced equation, 2 moles of AlCl₃ are produced for every 2 moles of aluminum that react, yielding 0.11 moles of AlCl₃.

$$0.11 \text{ mol Al} \times \frac{2 \text{ mol AlCl}_3}{2 \text{ mol Al}} = 0.11 \text{ mol AlCl}_3$$

Using the CuCl₂ solution **concentration** given in Table 1 to evaluate the other reactant shows that 500. mL of a 1.0 M solution of CuCl₂ is equivalent to 0.50 moles of CuCl₂.

$$500. \text{ mL} \times \frac{1 \text{ L}}{1,000 \text{ mL}} \times \frac{1.0 \text{ mol CuCl}_2}{1 \text{ L}} = 0.50 \text{ mol CuCl}_2$$

The balanced equation shows that for every 3 moles of CuCl₂ that react, 2 moles of AlCl₃ are produced, yielding 0.33 moles of AlCl₃.

$$0.50 \text{ mol CuCl}_2 \times \frac{2 \text{ mol AlCl}_3}{3 \text{ mol CuCl}_2} = 0.33 \text{ mol AlCl}_3$$

Because 500. mL of 1.0 M CuCl₂ theoretically produces 0.33 moles of AlCl₃ but 3.0 g of Al only produces 0.11 moles of AlCl₃, aluminum must be the limiting reactant because it produces the smallest amount of product (ie, 0.11 mol < 0.33 mol).

(Choices B and D) Copper and aluminum chloride are the *products* of the reaction, so neither of these can be the limiting reactant.

(Choice C) Based on the calculations, copper(II) chloride is in excess of aluminum, and therefore cannot be the limiting reactant.

Educational objective:

The limiting reactant is the reactant that is completely consumed first in a chemical reaction. It limits and determines the amount of product that can be produced based on the stoichiometric ratios of the chemical reaction.

Time spent:0 sec

Subject:
General Chemistry

QID:402069

System:
Interactions of Chemical Substances

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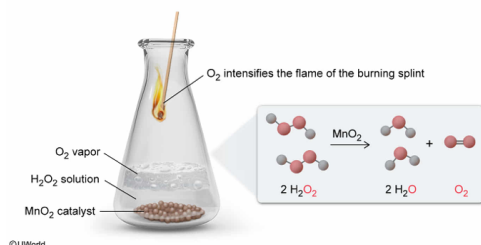
Based on the information in the passage, what is the identity of the gas formed from the decomposition of H_2O_2 ?

- ☐ A. Hydrogen [4%]
☐ B. Carbon dioxide [2%]
☐ C. Nitrogen
☒ D. Oxygen [92%]

Correct

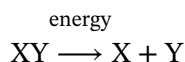
93% answered correctly

Explanation:



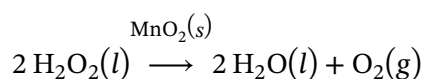
Chemical **reactions** are classified into **types** based on general patterns. This practice assists in the **prediction** of the **products** of a reaction that can then be verified with the correct tests in the laboratory.

A **decomposition** reaction follows the general pattern of one reactant breaking down into two or more products made from the elements of the original compound:



Decomposition reactions often require an energy source to break the bonds in the reactant. However, for the decomposition of H₂O₂, MnO₂ **catalyzes** the breaking of the hydrogen and oxygen bonds in peroxide by forming a reactive intermediate.

Because the glowing splint burned more brightly in the gas that was produced when MnO₂ was added, one of the products of the decomposition of H₂O₂ must be O₂. From this, it can be inferred that the other product is H₂O. Accordingly, the net reaction for the decomposition of H₂O₂ is:



(Choice A) Because the burning splint test did not make the sound of a "pop" (indicating the ignition of hydrogen gas), the evidence does not support that H₂ was present in the reaction flask.

(Choices B and C) Because there are no carbon or nitrogen atoms in H₂O₂, neither carbon dioxide (CO₂) nor nitrogen gas (N₂) can be products of the reaction. In addition, if CO₂ were present, the glowing splint test would have extinguished the embers.

Educational objective:

Reaction types follow a general pattern and allow for the prediction of the products of a reaction that can then be verified in the laboratory.

Time spent: 0 sec

Subject:
General Chemistry

QID: 402070

System:
Interactions of Chemical Substances

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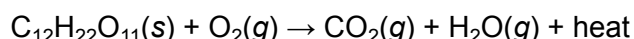
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Hydrogen peroxide: $\text{H}_2\text{O}_2(\text{aq})$	MnO_2 catalyst and H_2O_2 were combined in a tall flask	Burned more brightly when placed in the gas

When sugar is burned completely, the following unbalanced combustion reaction takes place:



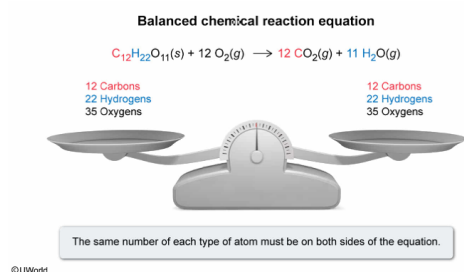
If the reaction is balanced, the ratio of $\text{CO}_2(\text{g})$ to $\text{H}_2\text{O}(\text{g})$ is:

- ☐ A. 1 to 1. [4%]
☐ B. 3 to 2. [3%]
☐ C. 6 to 5. [4%]
☒ D. 12 to 11. [88%]

Correct

87% answered correctly

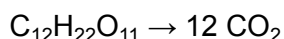
Explanation:



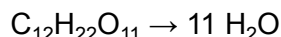
Reactions can be categorized by "type" based on certain reactants and reaction conditions. For example, a **combustion** reaction occurs when a hydrocarbon, alcohol, or another organic molecule reacts with oxygen (ie, burns) to produce **products** of CO_2 and H_2O .

Knowing reaction types helps predict the *products* of the reaction, but it does not predict the *amount* of each product that is produced. Instead, this is found by applying the **conservation of mass**, which requires that the number of a given type of atom must be the same on both sides of the chemical equation.

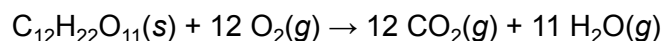
The **chemical equation** must be **balanced** to know the ratios of the **stoichiometric coefficients** of the products compared to the reactants. To find the stoichiometric ratios of the products from the combustion of sugar, balance the number of C atoms in sugar with CO_2 :



Next, balance the number of H atoms in sugar with H_2O :



Finally, there should be 12 O_2 molecules on the left side of the equation to balance the number of oxygen atoms on the right:



Therefore, the combustion of sugar produces 12 $\text{CO}_2(\text{g})$ and 11 $\text{H}_2\text{O}(\text{g})$.

(Choice A) This ratio gives only 1 carbon, 2 hydrogen, and 3 oxygen atoms on the right side of the equation.

(Choice B) This ratio gives only 3 carbon, 4 hydrogen, and 8 oxygen atoms on the right side of the equation.

(Choice C) This ratio gives only 6 carbon, 10 hydrogen, and 17 oxygen atoms on the right side of the equation.

Educational objective:

The conservation of mass dictates that the number of any type of atom be the same on both sides of a chemical equation (ie, the equation must be balanced).

Time spent: 0 sec
Subject:
General Chemistry

QID: 402071
System:
Interactions of Chemical Substances

Students were asked to bring items from home to serve as examples for different types of reactions. The instructor provided additional reactants, solvents, heat, or a catalyst to demonstrate reactions with the household items.

The students brought aluminum foil, table salt, sugar, baking soda, vinegar, toilet bowl cleaner, hydrogen peroxide, and antacid tablets.

Experiment 1

Reactions were performed using the household items. The reaction conditions and student observations are recorded in Table 1.

Table 1 Reaction Conditions and Visual Observations

Household item	Reaction conditions	Observations
Antacid: CaCO_3	Ground to powder and suspended in water; filtered to isolate CaCO_3 ; heat added	White solid (remained); some vapor observed
Sugar: $\text{C}_{12}\text{H}_{22}\text{O}_{11}$	Granular sugar was mixed with KClO_3 and lit with a match attached to the end of a long metal rod.	The sugar caught on fire and burned for several seconds
Aluminum foil: Al(s)	Placed in a solution of 1.0 M CuCl_2	Solution changed from blue-green to cloudy; red solid formed on the foil surface; temperature increased
Table salt: NaCl	Dissolved in water and then added to a solution of 0.05 M AgNO_3	White solid precipitate formed
Vinegar: $\text{HC}_2\text{H}_3\text{O}_2(\text{aq})$	Added baking soda (NaHCO_3) to 2 mL vinegar in a test tube	Fizzed and bubbled; produced a gas
Toilet bowl cleaner: HCl(aq)	Added a small piece of Al foil into a test tube containing 2 mL of cleaner	After several minutes, bubbles appeared; bubbles were produced until the Al foil disappeared
Hydrogen peroxide: $\text{H}_2\text{O}_2(\text{aq})$	Some MnO_2 catalyst was added to H_2O_2 in a tall flask	Bubbles formed on the surface of the liquid

Experiment 2

The reactions that were observed to produce gas bubbles in Experiment 1 were repeated, and samples of the gases produced were collected. Each gas sample was tested by holding a burning (glowing) wooden splint within the reaction vessel to determine if the reaction produced O_2 , H_2 , or CO_2 . A glowing splint placed in O_2 will burn more brightly, one placed in H_2 will ignite the gas and cause a loud noise, and one placed in CO_2 will be extinguished. The results of the gas tests are recorded in Table 2.

Table 2 Glowing Splint Gas Test Observations

Household item	Reaction conditions	Glowing splint observations
Antacid: CaCO_3	CaCO_3 powder heated in the bottom of a tall test tube	Dimmed and was extinguished when placed in the gas
Vinegar: $\text{HC}_2\text{H}_3\text{O}_2(\text{aq})$	NaHCO_3 and 2 mL vinegar in the bottom of a tall test tube	Dimmed and was extinguished when placed in the gas
Toilet bowl cleaner: HCl(aq)	Small piece of Al foil placed in 2 mL of cleaner in a tall test tube	Sound of a pop was heard when placed in the gas
Hydrogen peroxide: $\text{H}_2\text{O}_2(\text{aq})$	MnO_2 catalyst and H_2O_2 were combined in a tall flask	Burned more brightly when placed in the gas

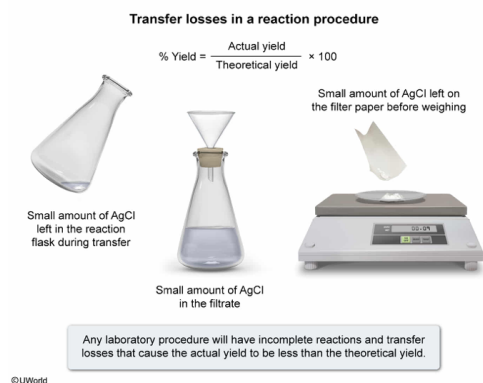
How will the percent yield of the reaction with table salt and AgNO_3 be affected if some of the solid precipitate is lost when it is filtered from the solution?

- ☐ A. The percent yield would be greater than 100% because the mass of the isolated product is greater than the calculated mass. [1%]
- ☐ B. The percent yield would be greater than 100% because the mass of the isolated product is less than the calculated mass. [1%]
- ☐ C. The percent yield would be less than 100% because the mass of the isolated product is greater than the calculated mass. [3%]
- ✔ ☒ D. The percent yield would be less than 100% because the mass of the isolated product is less than the calculated mass. [93%]

Correct

94% answered correctly

Explanation:



The **percent yield** of a chemical reaction assesses the efficiency of a reaction and the laboratory procedure to isolate the product. Percent yield is the **ratio** of the amount of the product obtained (the **actual yield**) to the calculated **theoretical yield**.

$$\% \text{ Yield} = \frac{\text{Actual yield}}{\text{Theoretical yield}} \times 100$$

The theoretical yield of a chemical reaction is the maximum amount of product possible based on stoichiometric calculations. The value is based on the number of moles of the limiting reactant and the stoichiometric ratios in the balanced chemical reaction.

The actual yield obtained after conducting a reaction will be *less* than the theoretical yield when some of the product is lost in the process of executing laboratory procedures such as filtering or transferring. The actual yield could appear to be *greater* than the theoretical yield when a product contains impurities or has not been completely dried before weighing.

In the reaction between NaCl (table salt) and AgNO₃, solid AgCl is produced. If some of the solid AgCl is lost during filtration to isolate the product, the actual mass of the product obtained will be less than the calculated theoretical mass. According to the equation, if the actual yield is less than the theoretical yield, then the percent yield will be less than 100%.

(Choices A and B) Losing mass during filtration would cause the actual yield to be *less* than the theoretical yield. According to the equation for percent yield, this would cause the percent yield to be *less than* 100%, not greater.

(Choice C) If the mass of the isolated product was apparently greater than the calculated mass because the isolated mass was insufficiently dried (ie, had extra mass from water), it would mathematically give a percentage *greater than* (not less than) 100%.

Educational objective:

The percent yield is a description of the efficiency of a reaction. It is the ratio of the mass obtained from an experiment (the actual yield) to the calculated mass (theoretical yield) expressed as a percentage.

Time spent: 0 sec

Subject:
General Chemistry

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System:
Interactions of Chemical Substances